ETIOLOGY AND PATHOGENESIS

Although many theories abound, the cause of seborrheic dermatitis remains unknown.

SEBORRHEA

Seborrhea is associated with oily-looking skin (seborrhea oleosa), although increased sebum production cannot always be detected in these patients. Even if seborrhea provides a predisposition, seborrheic dermatitis is not primarily a disease of the sebaceous glands.

The high incidence of seborrheic dermatitis in newborns parallels the size and activity of the sebaceous glands at this age. Newborns have large sebaceous glands with high sebum secretion rates, similar to adults. In childhood, sebum production and seborrheic dermatitis are closely linked. However, in adulthood, this association weakens—sebaceous gland activity peaks in early puberty, yet seborrheic dermatitis may develop decades later.

The sites most commonly affected—face, ears, scalp, and upper trunk—are rich in sebaceous follicles. Two conditions frequently occur in these areas: seborrheic dermatitis and acne. Histologically, sebaceous glands in patients with seborrheic dermatitis are often enlarged on cross-section. One study found that while total skin surface lipids were not elevated, the lipid composition showed an increased proportion of cholesterol, triglycerides, and paraffin, and a decrease in squalene, free fatty acids, and wax esters. These mild lipid abnormalities may result from impaired keratinization, which is often evident on histopathology.

Seborrheic dermatitis is more common in patients with parkinsonism and other neurologic disorders, in whom sebum secretion is increased. Similarly, when sebum production is reduced by medications such as levodopa or promestriene, seborrheic dermatitis may improve.

The term *eczema flannelaire* originates from the belief that retention of skin surface lipids due to clothing—particularly rubbing from rough textiles like flannel or synthetic underwear—may promote or worsen seborrheic dermatitis.

MICROBIAL EFFECTS

Unna and Sabouraud, among the first to describe the condition, proposed a bacterial, yeast, or combined microbial etiology. This hypothesis has not been definitively proven, although bacteria and yeasts are present in large numbers on affected skin.

In infants, *Candida albicans* is frequently isolated from dermatitic skin lesions and stool samples. Sensitization to *C. albicans* has been demonstrated through positive intracutaneous candidin tests, serum agglutinating antibodies, and lymphocyte transformation tests. However, a direct pathogenic role for *C. albicans* in seborrheic dermatitis remains unproven.

Aerobic bacterial counts on the scalp were lower in patients with seborrheic dermatitis (140,000 bacteria/cm²) compared to unaffected individuals (280,000 bacteria/cm²) and those with dandruff (250,000 bacteria/cm²). In contrast, *Staphylococcus aureus* was rarely found in unaffected persons or those with dandruff but was isolated in approximately 20% of seborrheic dermatitis patients, accounting for about 32% of the total skin flora.

Propionibacterium acnes counts were reduced in patients with seborrheic dermatitis (7,550 bacteria/cm² in those without dandruff).

SEBORRHEIC DERMATITIS

AT A GLANCE

- Infantile and adult forms exist.
- Characterized by erythema and greasy scaling.

- Lesions commonly affect the scalp, ears, face, chest, and intertriginous areas.
- Generalized and even erythrodermic forms may occur.
- Etiology unknown but may involve increased sebum secretion, altered sebum composition, certain drugs, or *Malassezia* yeasts.
- May serve as a cutaneous marker of HIV infection and AIDS, especially when severe, atypical, or resistant to therapy.